Exponentials & Logarithms

Section 1: Exponential Equations

1 Solve for x: $3^{2x-1} = 27$.

a. $x = 2$

b. $x = 3$

c. $x = 1$

d. $x = 4$

2 If $4^{x+1} = \left(\frac{1}{2}\right)^{x-4}$, what is x?

a. $x = 2$

b. $x = 1$

c. $x = 3$

d. $x = -2$

3 What is the horizontal asymptote of the exponential function $f(x) = 2^x - 5$?

a. $y = -5$

b. $y = 5$

c. $y = 0$

d. $x = -5$

Section 2: Logarithmic Properties

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4 What is the value of $\log_2(32)$?

a. 5

b. 4

c. 6

d. 16

5 Evaluate $\log_3(54) - \log_3(2)$.

a. 3

b. 2

c. 4

d. 27

6 Simplify: $2\log_5(x) + \log_5(y)$.

a. $\log_5(x^2y)$

b. $\log_5(2xy)$

c. $\log_5(x^2 + y)$

d. $\log_5(xy^2)$

Section 3: Logarithmic Equations

7 Solve for x: $\log_4(2x + 8) = 3$.

a. $x = 28$

b. $x = 24$

c. $x = 32$

d. $x = 16$

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8 Solve for x: $\log_2(x) + \log_2(x-2) = 3$.

a. $x = 4$

b. $x = 2$

c. $x = -2$

d. $x = 6$

9 What is the domain of the function $f(x) = \log_3(4 - x)$?

a. $(-\infty, 4)$

b. $[4, \infty)$

c. $(4, \infty)$

d. $(-\infty, 4]$

Section 4: Natural Logarithms & Base e

10 Evaluate $\ln(e^7)$.

a. 7

b. e

c. 1

d. 14

11 Solve the natural exponential equation: $e^{2x} = 5$.

a. $x = \frac{\ln 5}{2}$

b. $x = 2\ln 5$

c. $x = \ln(2.5)$

d. $x = \frac{5}{2e}$

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12 What is the value of $e^{\ln(12)}$?

a. 12

b. 1

c. e

d. 144

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